1 Data Structures for Disjoint Sets

Disjoint sets (Union-Find)

Each set identified by a representative.

Operations

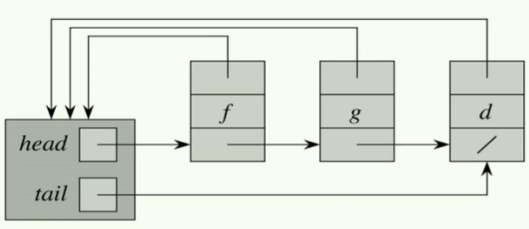
- `MAKE-SET(x)`: $S_i = \{x\}$, add to $\mathcal{S}$

Let $x \in S_x$, $y \in S_y$

- `UNION(x, y)`: $\mathcal{S} = \mathcal{S} \setminus \{S_x, S_y\} \cup \{S_x \cup S_y\}$
- `FIND(x)`: return representative of x in S_x

```
CONNECTED-COMPONENTS(G)
    for each vertex v in G.V
        MAKE-SET(v)
    for each edge (u, v) in G.E
        if FIND(u) != FIND(v)
            UNION(u, v)
```

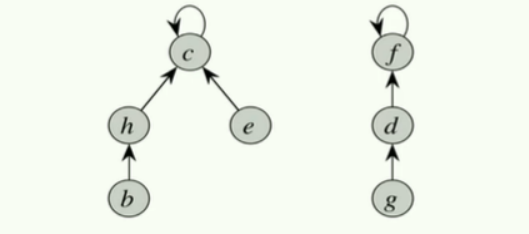
Linked list representation



Weighted-union heuristic

Always merge smaller into larger. m ops on n elements: $O(m + n \lg n)$. Each element's representative updated $\leq \lg n$ times (each update at least doubles set size).

Tree representation

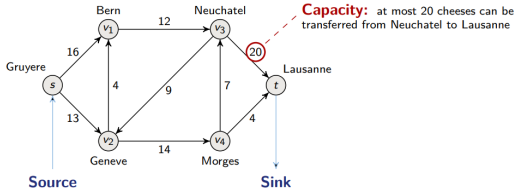


Runtime with both heuristics

Union by rank: smaller-rank root → child of larger-rank root (rank ≈ height upper bound).
Path compression: all nodes on FIND path become direct children of root.
Union by rank + path compression: $O(m \cdot \alpha(n))$ for m operations on n elements. $\alpha(n) \leq 5$ for any practical n. Bound is tight.

2 Flow network

Definitions



Formally

- Source: $s \in V$, sink: $t \in V$
- Directed graph (DG): $G = (V, E)$
- Each (u, v) has a capacity $c(u, v) \geq 0$
- No antiparallel: if $(u, v) \in E$, then $(v, u) \notin E$

Capacity constraint

$\forall u, v \in V,\ 0 \leq f(u, v) \leq c(u, v)$

Flow conservation

$\forall u \in V \setminus \{s, t\},\ \sum_{v \in V} f(u, v) = \sum_{v \in V} f(v, u)$

Cut

Partition of V into two disjoint sets S and T such that $s \in S$ and $t \in T$. $c(S, T) = \sum_{u \in S, v \in T} c(u, v)$.

Maximum flow problem - Ford-Fulkerson method

Residual network $G_f = (V, E_f)$

$c_f(u, v)$: additional flow possible (forward edge) or removable (backward edge).

```
FORD-FULKERSON-METHOD(G, s, t)
    initialize flow f to 0
    while exists an augmenting path p in G_f
        b = bottleneck(p)
        augment flow along p by b
    return f
```

Min cut problem

Min cut

It is a cut where the capacity crossing it is the minimum among all cuts.

Net flow accross cut

$f(S, T) = \sum_{u \in S, v \in T} f(u, v) - \sum_{u \in S, v \in T} f(v, u)$

Max-flow min-cut theorem (⇔)

- f is the maximum flow
- G_f has no augmenting path
- $|f| = c(S, T)$ for a minimum cut (S, T)

Applications of max flow

Bipartite matching as flow

$G = (V, E)$, $V = L \cup R$. Connect $s \to \forall u \in L$ and $\forall v \in R \to t$, all capacities 1. Run Ford-Fulkerson.

Edge-disjoint paths

Max # edge-disjoint paths from s to t = min edges to remove to disconnect s from t = max flow (all capacities 1).

3 Spanning Trees

Definition

Set T of edges that is acyclic and spanning (connects all vertices of G).

Minimum Spanning Trees (MST) problem

In: undirected $G = (V, E)$, weights $w(u, v)$ for each $(u, v) \in E$.
Out: spanning tree of minimum total weight.

Cut property (safe edge)

Let $A \subseteq T_{\text{MST}}$, $(S, V \setminus S)$ a cut respecting A, and (u, v) a light (min-weight) edge crossing the cut. Then (u, v) is *safe* to add to A.

Prim's algorithm

```
PRIM(G, w, r)                    // w: E -> R weights
    Q = {}          // min-priority queue (min-heap)
    for each u in G.V:
        u.key = inf
        u.pi = NIL               // u.pi = parent in T
        INSERT(Q, u)
    DECREASE-KEY(Q, r, 0)                // r.key = 0
    while Q != {}:
        u = EXTRACT-MIN(Q)
        for each v in G.Adj[u]:
            if v in Q and w(u,v) < v.key:
                v.pi = u
                DECREASE-KEY(Q, v, w(u,v))
```

Why does it work?

T is always a subtree of a MST.

Prim's runtime

$O(E \lg V)$ with binary heap. $O(V \lg V + E)$ with Fibonacci heap.

4 ————————

Key definiti2ons

Linear map

$T : V \to W$ is linear if $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$ for all $x, y \in V$, $\alpha, \beta \in \mathbb{R}$.

Rank–nullity

For $T : V \to W$ with $\dim V < \infty$:

$$\dim \ker T + \dim \operatorname{im} T \;=\; \dim V.$$

Formulas to remember

- **Quadratic:** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Euler:** $e^{i\theta} = \cos\theta + i \sin\theta$
- **Geom. sum:** $\sum_{k=0}^{n} r^k = \dfrac{1 - r^{n+1}}{1 - r}$, $r \neq 1$
- **Taylor:** $f(x) = \sum_{k \geq 0} \dfrac{f^{(k)}(a)}{k!}(x - a)^k$

Pitfall

$\log(ab) = \log a + \log b$ *only* when $a, b > 0$.

Useful identities

$$\sin^2\theta + \cos^2\theta = 1$$
$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$
$$\sin(2\theta) = 2\sin\theta\cos\theta$$

5 Topic 2 — Probability

Basics

- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
- Bayes: $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$

Common distributions

Dist.	PMF/PDF	$\mathbb{E}[X]$
Bern(p)	$p^x(1-p)^{1-x}$	p
Bin(n,p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np
Poiss(λ)	$e^{-\lambda}\lambda^x/x!$	λ
$\mathcal{N}(\mu,\sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$	μ

Worked

$X \sim \mathcal{N}(0,1)$. Then $\mathbb{P}(|X| < 1) \approx 0.6827$.

6 Topic 3 — Algorithms

Complexity cheat

Op	Cost
Sort (comparison)	$\Theta(n \log n)$
Hash lookup (avg)	$\Theta(1)$
BFS / DFS	$\Theta(V + E)$
Dijkstra (heap)	$\Theta((V + E)\log V)$

Code snippet

```
def bsearch(a, x):
    lo, hi = 0, len(a)-1
    while lo <= hi:
        m = (lo+hi)//2
        if a[m] == x: return m
        if a[m] <  x: lo = m+1
        else:         hi = m-1
    return -1
```

7 Quick reference

Greek: $\alpha\,\beta\,\gamma\,\delta\,\varepsilon\,\zeta\,\eta\,\theta\,\iota\,\kappa\,\lambda\,\mu\,\nu\,\xi\,\pi\,\rho\,\sigma\,\tau\,\phi\,\chi\,\psi\,\omega$

Sets: $\mathbb{R}\;\mathbb{N}\;\mathbb{Z}\;\mathbb{Q}\;\mathbb{C}$ **Logic:** $\forall\;\exists\;\Rightarrow\;\Leftrightarrow\;\neg$

Derivatives: $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\ln x)' = 1/x$, $(e^x)' = e^x$, $(x^n)' = nx^{n-1}$.

Integrals: $\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1}$, $\int \frac{1}{x}\,\mathrm{d}x = \ln|x|$, $\int e^x\,\mathrm{d}x = e^x$.

8 ————————

Key definitions

Linear map

$T : V \to W$ is linear if $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$ for all $x, y \in V$, $\alpha, \beta \in \mathbb{R}$.

Rank–nullity

For $T : V \to W$ with $\dim V < \infty$:

$$\dim \ker T + \dim \operatorname{im} T \;=\; \dim V.$$

Formulas to remember

- **Quadratic:** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Euler:** $e^{i\theta} = \cos\theta + i \sin\theta$
- **Geom. sum:** $\sum_{k=0}^{n} r^k = \dfrac{1 - r^{n+1}}{1 - r}$, $r \neq 1$
- **Taylor:** $f(x) = \sum_{k \geq 0} \dfrac{f^{(k)}(a)}{k!}(x - a)^k$

Pitfall

$\log(ab) = \log a + \log b$ *only* when $a, b > 0$.

Useful identities

$$\sin^2\theta + \cos^2\theta = 1$$
$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$
$$\sin(2\theta) = 2\sin\theta\cos\theta$$

9 Topic 2 — Probability

Basics

- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
- Bayes: $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$

Common distributions

Dist.	PMF/PDF	$\mathbb{E}[X]$
Bern(p)	$p^x(1-p)^{1-x}$	p
Bin(n,p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np
Poiss(λ)	$e^{-\lambda}\lambda^x/x!$	λ
$\mathcal{N}(\mu,\sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$	μ

Worked

$X \sim \mathcal{N}(0,1)$. Then $\mathbb{P}(|X| < 1) \approx 0.6827$.

10 Topic 3 — Algorithms

Complexity cheat

Op	Cost
Sort (comparison)	$\Theta(n \log n)$
Hash lookup (avg)	$\Theta(1)$
BFS / DFS	$\Theta(V + E)$
Dijkstra (heap)	$\Theta((V + E)\log V)$

Code snippet

```
def bsearch(a, x):
    lo, hi = 0, len(a)-1
    while lo <= hi:
        m = (lo+hi)//2
        if a[m] == x: return m
        if a[m] <  x: lo = m+1
        else:         hi = m-1
    return -1
```

11 Quick reference

Greek: $\alpha\,\beta\,\gamma\,\delta\,\varepsilon\,\zeta\,\eta\,\theta\,\iota\,\kappa\,\lambda\,\mu\,\nu\,\xi\,\pi\,\rho\,\sigma\,\tau\,\phi\,\chi\,\psi\,\omega$

Sets: $\mathbb{R}\;\mathbb{N}\;\mathbb{Z}\;\mathbb{Q}\;\mathbb{C}$ **Logic:** $\forall\;\exists\;\Rightarrow\;\Leftrightarrow\;\neg$

Derivatives: $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\ln x)' = 1/x$, $(e^x)' = e^x$, $(x^n)' = nx^{n-1}$.

Integrals: $\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1}$, $\int \frac{1}{x}\,\mathrm{d}x = \ln|x|$, $\int e^x\,\mathrm{d}x = e^x$.

12 ————————

Rank–nullity

For $T : V \to W$ with $\dim V < \infty$:

$$\dim \ker T + \dim \operatorname{im} T \;=\; \dim V.$$

Formulas to remember

- **Quadratic:** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Euler:** $e^{i\theta} = \cos\theta + i \sin\theta$
- **Geom. sum:** $\sum_{k=0}^{n} r^k = \dfrac{1 - r^{n+1}}{1 - r}$, $r \neq 1$
- **Taylor:** $f(x) = \sum_{k \geq 0} \dfrac{f^{(k)}(a)}{k!}(x - a)^k$

Pitfall

$\log(ab) = \log a + \log b$ *only* when $a, b > 0$.

Useful identities

$$\sin^2\theta + \cos^2\theta = 1$$
$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$
$$\sin(2\theta) = 2\sin\theta\cos\theta$$

13 Topic 2 — Probability

Basics

- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
- Bayes: $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$

Common distributions

Dist.	PMF/PDF	$\mathbb{E}[X]$
Bern(p)	$p^x(1-p)^{1-x}$	p
Bin(n,p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np
Poiss(λ)	$e^{-\lambda}\lambda^x/x!$	λ
$\mathcal{N}(\mu,\sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$	μ

Worked

$X \sim \mathcal{N}(0,1)$. Then $\mathbb{P}(|X| < 1) \approx 0.6827$.

14 Topic 3 — Algorithms

Complexity cheat

Op	Cost
Sort (comparison)	$\Theta(n \log n)$
Hash lookup (avg)	$\Theta(1)$
BFS / DFS	$\Theta(V + E)$
Dijkstra (heap)	$\Theta((V + E)\log V)$

Code snippet

```
def bsearch(a, x):
    lo, hi = 0, len(a)-1
    while lo <= hi:
        m = (lo+hi)//2
        if a[m] == x: return m
        if a[m] <  x: lo = m+1
        else:         hi = m-1
    return -1
```

15 Quick2 reference

Greek: $\alpha\,\beta\,\gamma\,\delta\,\varepsilon\,\zeta\,\eta\,\theta\,\iota\,\kappa\,\lambda\,\mu\,\nu\,\xi\,\pi\,\rho\,\sigma\,\tau\,\phi\,\chi\,\psi\,\omega$

Sets: $\mathbb{R}\;\mathbb{N}\;\mathbb{Z}\;\mathbb{Q}\;\mathbb{C}$ **Logic:** $\forall\;\exists\;\Rightarrow\;\Leftrightarrow\;\neg$

Derivatives: $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\ln x)' = 1/x$, $(e^x)' = e^x$, $(x^n)' = nx^{n-1}$.

Integrals: $\int x^n \, \mathrm{d}x = \frac{x^{n+1}}{n+1}$, $\int \frac{1}{x} \, \mathrm{d}x = \ln|x|$, $\int e^x \, \mathrm{d}x = e^x$.